Hempcrete Catalogue

A design and fabrication study exploring hempcrete as a modular building material, from growing hemp and mixing hurds, lime binder and water, through to a bonded-brick wall, floor and roof catalogue for a low-carbon housing unit.



Material testing, form finding and performative testing of hempcrete brick units informed a full construction catalogue and assembly sequence, from structural timber frame through to bricklaying, lime render and green roof finish.

01/09/2019 - date started

01/02/2020 - date ended

Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
eg BSc Architecuture and enviromental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

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Https://www.instagram.com/sarc_studio/



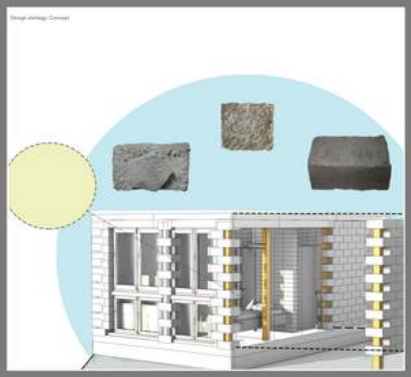
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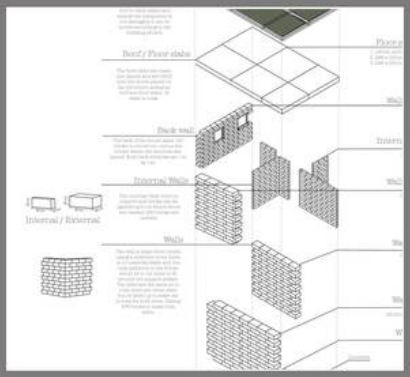
Hempcrete Catalogue

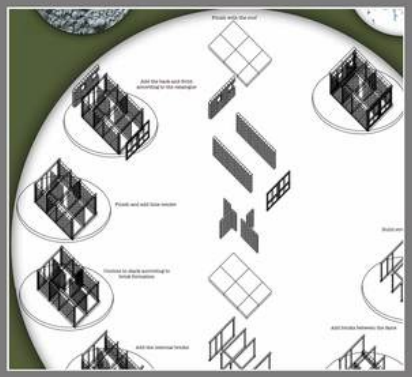
Modular Brick Fabrication & Assembly System

01/02/2020 - date ended

This portfolio documents the design strategy, fabrication process and construction system for a modular hempcrete building. Hemp hurds are mixed with a lime binder and water to cast bricks, which are tested for material performance and used to develop wall, floor and roof geometries. The document sets out a full construction catalogue, detailing wall types, window and louvre assemblies, floor slabs and green roof buildup, together with a step-by-step assembly sequence, from erecting the structural timber frame through to bricklaying, lime rendering and finishing. The result is a low-carbon, low-waste housing system built from a fast-growing natural material.







Sebastian Victor Dawber

W1610495

DES3B portfolio

Hemp hurds (3)



Lime binder (1)



Water (1)



Mix: hurd, binder .



Mix: Add water.



Mix: stir until even



Create mould box



Pour into box



Leave it to set



Final outcome



Final outcome



Final outcome



Design startegy: Concept....3

Design startegy: Hemp as a plant....4

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Geometry and digital form finding: Form finding....8

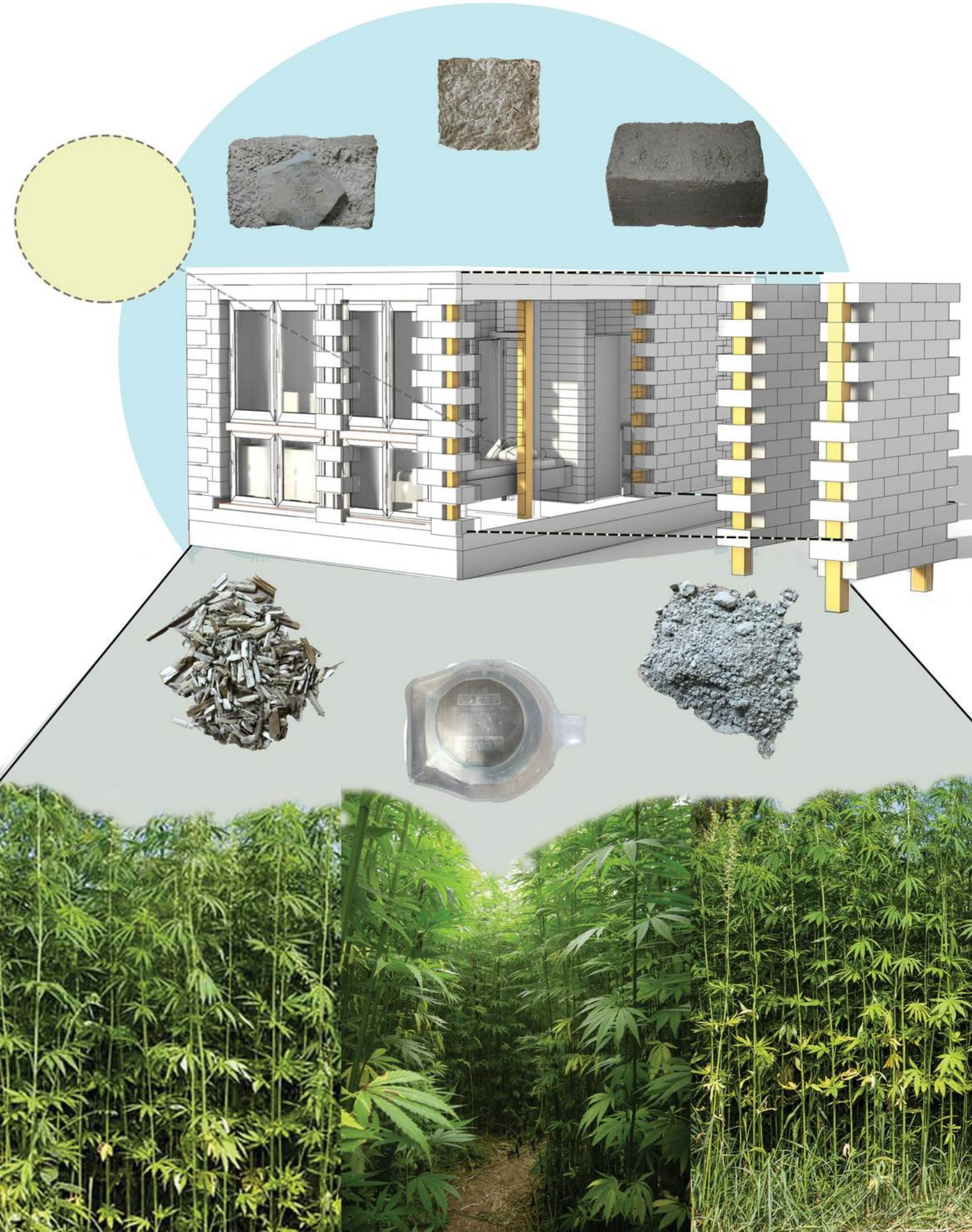
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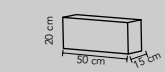
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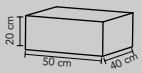


single-family house 2005

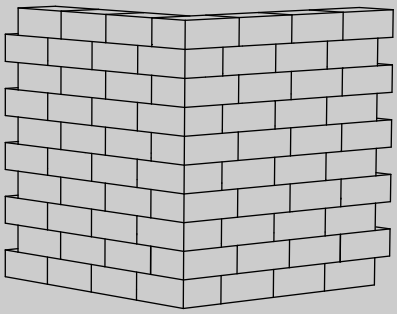




Thickness: 15cm	
Density (kg m/3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.53
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	36.55



Thickness: 40cm	
Density (kg m/3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96

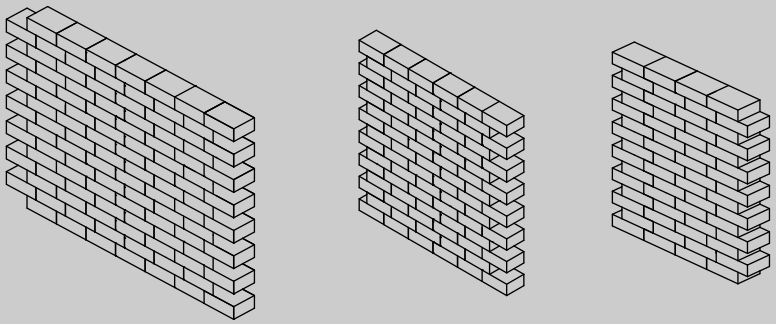


Cannabric, hempbricks. Compressed earth brick with hemp for insulating, loadbearing inside- and outside walls.

A nautural plant based material system using inudstrial hemp, when combined with a nautral binder and water forms into bulding compostiw material. It can used used as an extire wall with no extra insulation of internal parts needed, a vapour permeable lime render is used on the external side as the material is non-water resistant.

290 Mulberry / SHoP Architects





Due to struct building codes and the inability to build in the cofniend area SHoP archtiects had to create a system that could be assbmled while abding by the cityies strict bulding codes, due to the nature of the space conustrction work had to be minimizsed.

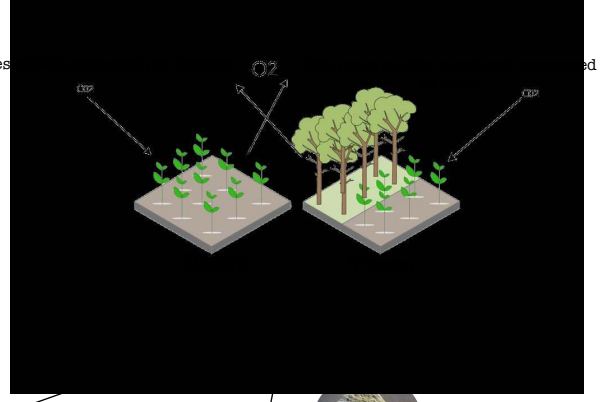
Making a literal interpretation of code written for classical ornamentation allowed us to project our enclosure over the property line at 10% intervals for every 100 square feet. Maximizing the amount of projected area, while minimizing the overall depth of the enclosure became key criteria for our design. When coupled with material properties and fabrication constraints, these criteria began to define an approach that was a contemporary reinterpretation of brick detailing. By customizing a standard precast brick panel system, we were able to achieve maximum effect at minimum cost. The building then becomes veiled by an textured wrapper around the street walls in contrast with the simplicity of the inner core.

Materiality and physical study model
Hemp plant anatomy

Design startegy: Hemp as a plant

Hemp

15 tonnes





Seeds

Oil
cooking oil
diet supplement
beauty products
fuel
paint

Hemp nut
milk
bakery
granola
protien powder

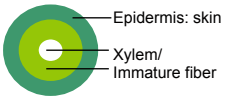
Stalk

Fiber
Texttiles
Insulation
Rope

Hurds
Paper
Compost
Animla bedding
Fiber board
Hempcrete
Stucco
Mortar

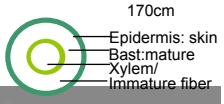
4.5m

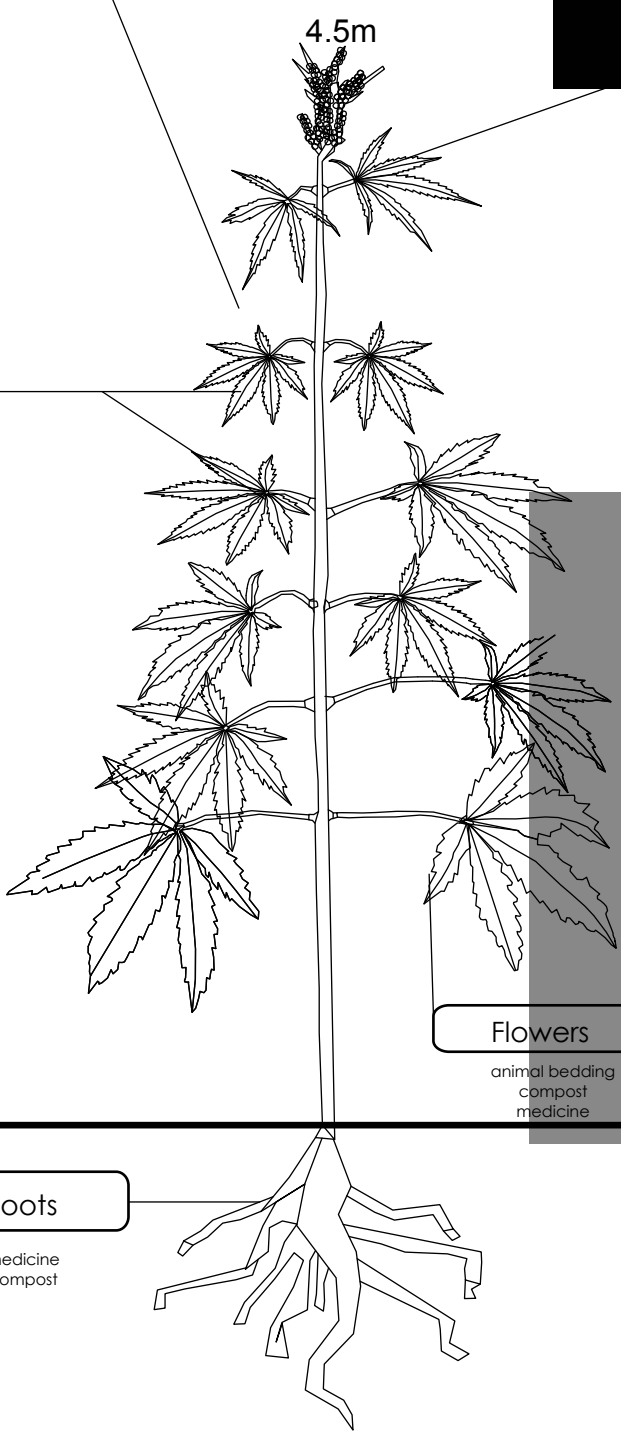


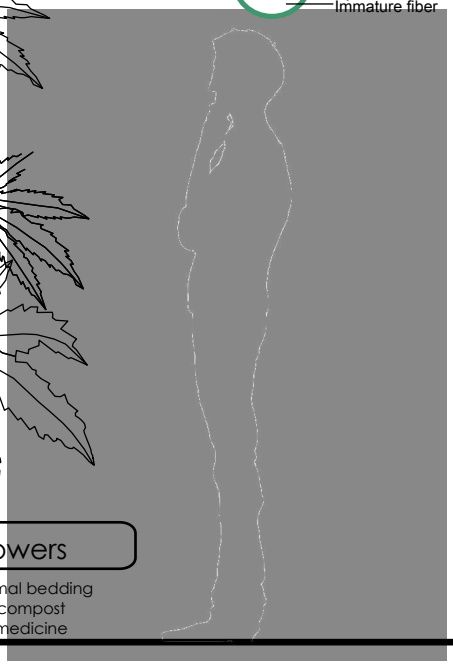




5.5ft
170cm







Flowers
animal bedding
compost
medicine

Roots
medicine
compost

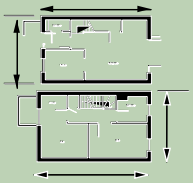
Hemp for hemp hurd extraction can be harvested in the vegatative stage

SEED	SEEDING	VEGETATIVE	FLOWERING	HARVEST
3 day	2 week	4-8 weeks	4-8 weeks	Varied

1 Precedent study







Precedents were chosen to give the design a foundation of ideas, a base to start the design.

A precedent from DES3A was integrated as it gave dimensions of a house made from hemp, a link between the two projects but the housing links the project, the social housing project.

Canabric was used to see how hemp bricks can be assembled.

290 Mulberry allowed me to see how brick panvels could be assembled and constructed in a confined space

2 Material testing

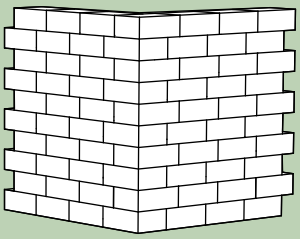




Material testing assessed the best ratio needed to create hemp bricks.

Hemp:Binder:Water – 2:1:1

3 Formation



+

Rhinoceros

Due to COVID-19 tests were limited and digital forms had to be constructed and chosen with little physical testing but more so online and digital research was used to find the final form. Using a stretcher brick bond and dimensions of 8mx5m.

4 Simulation

IES INTEGRATED ENVIRONMENTAL SOLUTIONS

Once the form was found IES VE was used to get accurate environmental simulations. Allowing me to see that hempcrete works better with closed rooms rather than open spaces.

5 Construction

Rhinoceros

The simulation gave a final form with was created and edited on Rhinoceros 3D to construct windows and created the finished panels as well as form.

6 Layout

Graphic

Graphic was used to lay out and present the final work.

Geometry and digital form finding: Material testing

"Ingredients" "Mix:Biner-Hurd-Water-Hurd" "Add Mix to box" "Test"

1.Binder (B)



2.Hemp hurds(H)



3.Water(W)



4.Mix: Binder-Hurds



5.Mix: Binder-Hurds-Water



7. Add rest of the hurds as you stir



8.Mix evenly



9.Pour mixture into mould box



10.Let it set for 48hrs



11.Create a ball



12.Poke a hole



17.If done correctly it sticks together



B:H:W- 1:3:1







To many hurds were added.
Bricks started breaking

B:H:W- 2:3:1









To mcuh binder was used

B:H:W- 1:2:1





-Uneven mix

B:H:W- 1:2:1



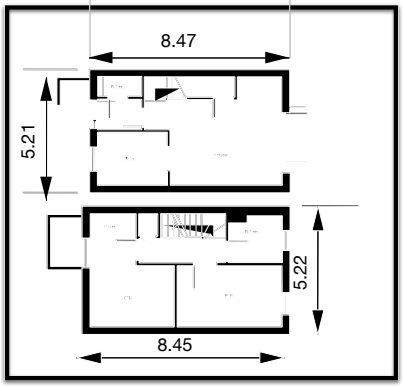


-Even mix

Perfect mix but mixing needs
to be done corretly

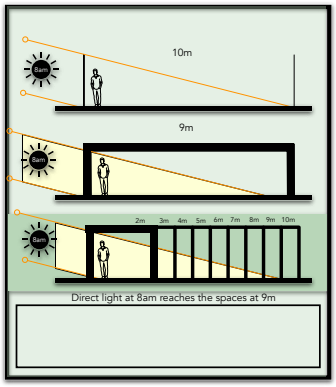
Adding a mix ratio of B:H:W- 1:2:1 leads to a successful hempcrete mixture

1 .Plans taken from Triangle Swindon, Glenn Howells architects. It gave dimensions for hempcrete housing, due to the house only having space for one person it was reduced to one floor with dimensions: 8.5 X 5.2m.



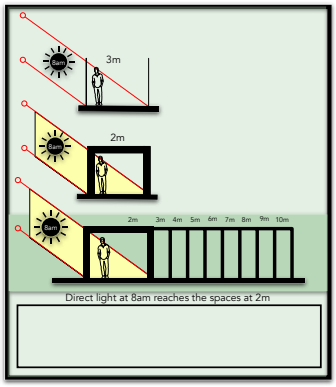
2. Sections were used to find the best length that will allow sun to enter the space at 8am, the time when residents wake up for work very morning.

22 March- Equinox solstice



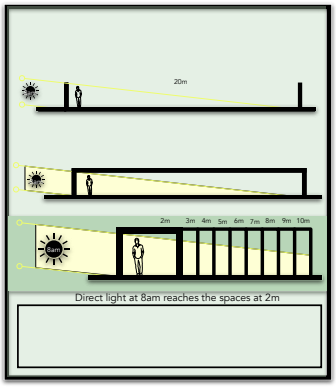
Direct light at 8am reaches the spaces at 9m

22 June- Summer solstice



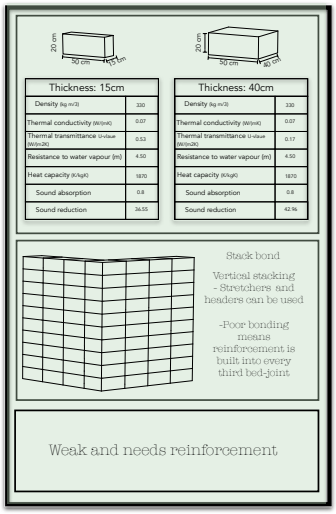
Direct light at 8am reaches the spaces at 2m

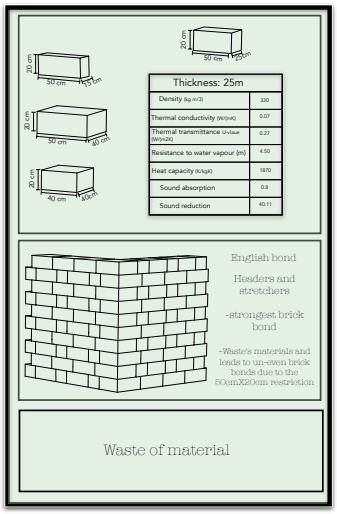
22 December- Wintersolstice

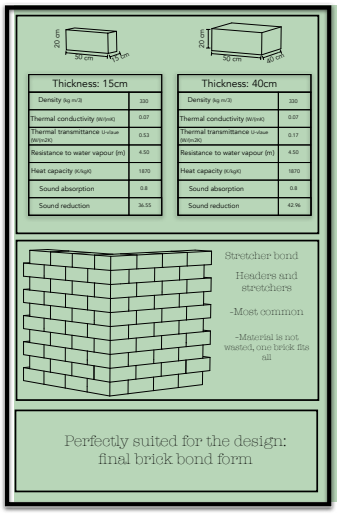


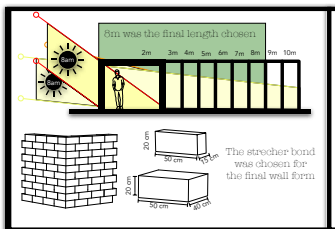
Direct light at 8am reaches the spaces at 2m

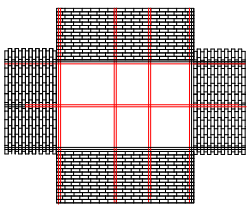
3. Three different wall bonding techqniues were exmained but due to COVID-19 testing was limited and a online research had to be used.







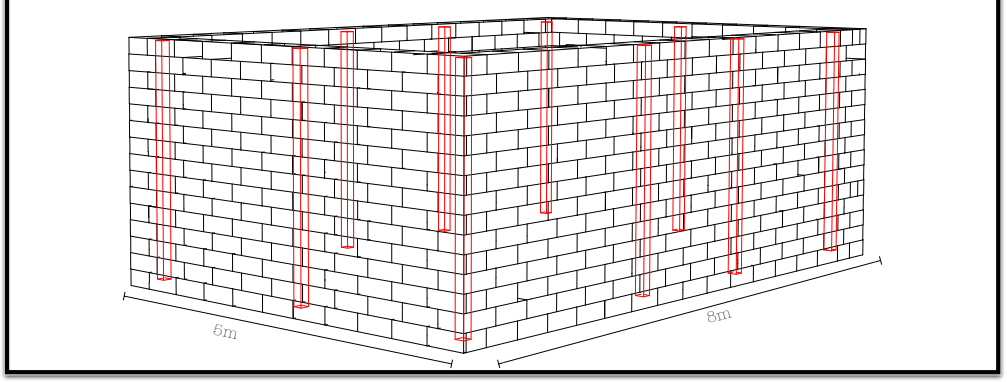


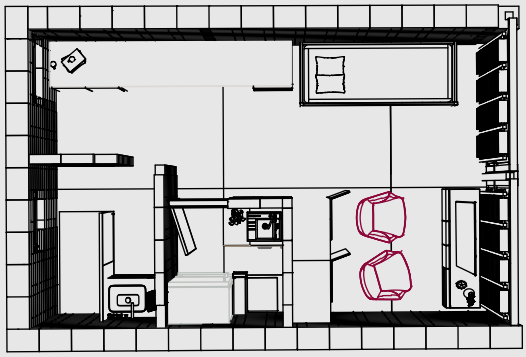


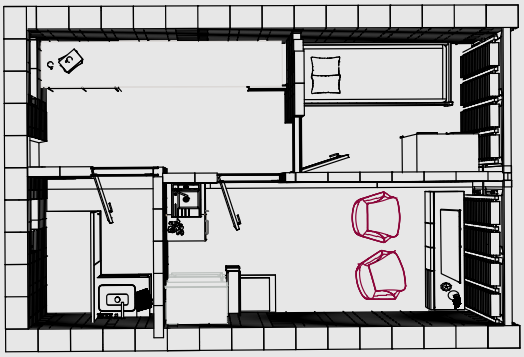
Final form

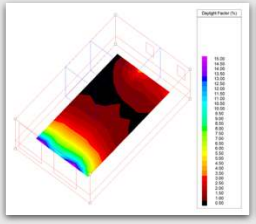
The final layout was developed using lines to place support pillars creating a catalouge of components

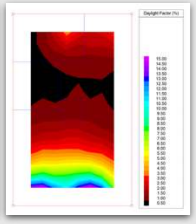
Using stretcher bond and a 5mx8m dimension similar to the Triangle Swindon, Glenn Howells architect precedent the social housing space was formed. Creating a single unit housing module that allows light to enter the main room at 8am every morning throughout the year, except for winter where it only enters at 9am

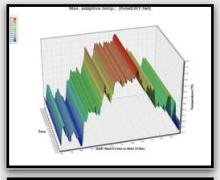


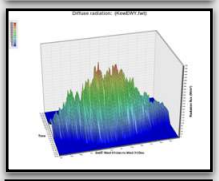


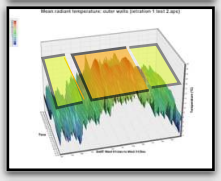


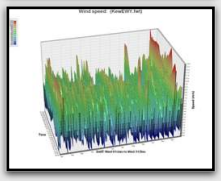




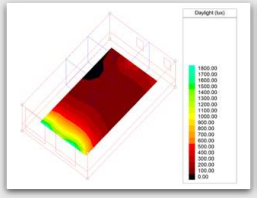


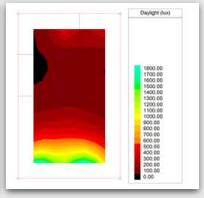




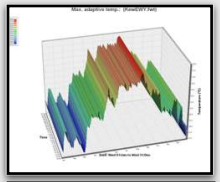


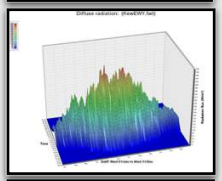
The high daylight factor and LUX level indicates the room is well lit but can be seen as too bright in areas. Light travels clearly through the entire space

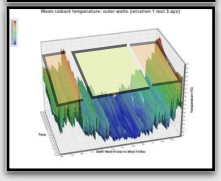


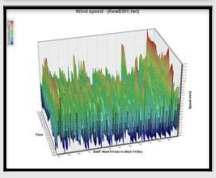


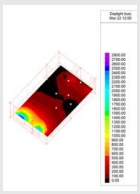
The first design uses an open floor plan to allow for a bigger space. The mean radiant temperature showed an interesting outcome and the open space is not as good. Higher temperatures in summer and lower temperatures in winter, this result is not describable but shows clear zoom for improvement.

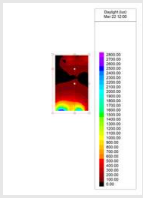




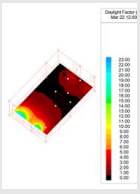


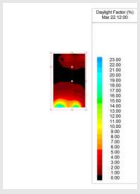




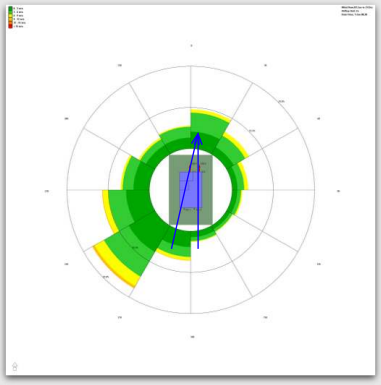


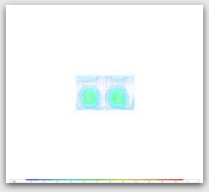
The daylight factor and LUX level indicates the room is well lit but has issues in areas that aren't close to the windows. Light travels through the space but doesn't perform as well as number 1.

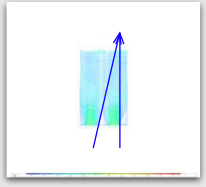


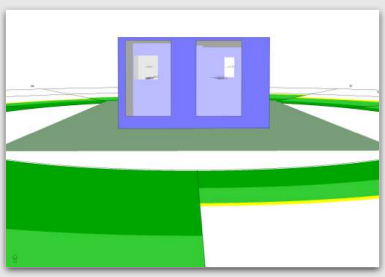


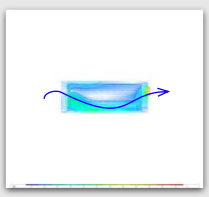
The second design uses a closed floor plan to allow for a confined spaces. This space had different results with a high mean radiant temperature in winter and low mean radiant temperature in summer which is preferable in the United Kingdom. Showing that hempcrete performs better within confined spaces regulating internal conditions effectively due to its hygroscopic nature.

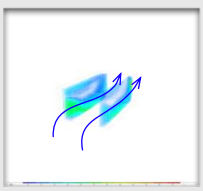




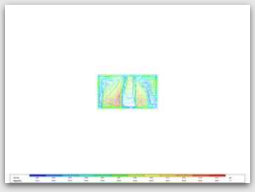


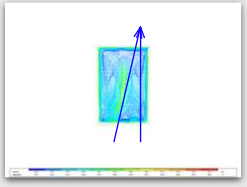


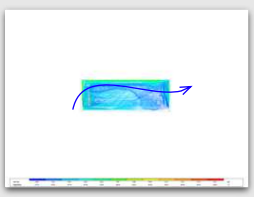




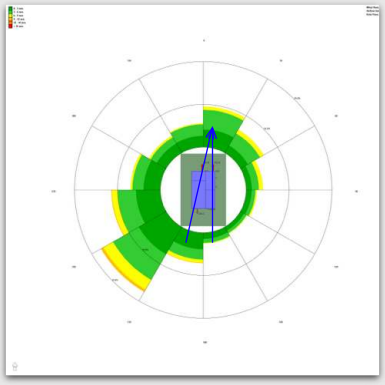
Adding windows on either side allows for cross ventilation. The open space had lower wind levels with wind being dispersed due to the open space nature. The wind travels from the floor and goes up, this may cause issues when walking around of working due to the wind currents coming from the floor.

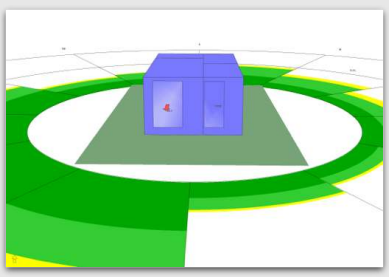




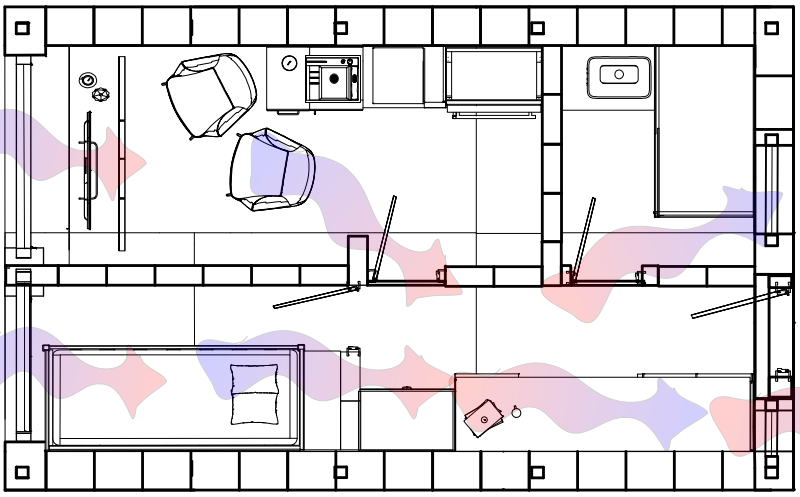


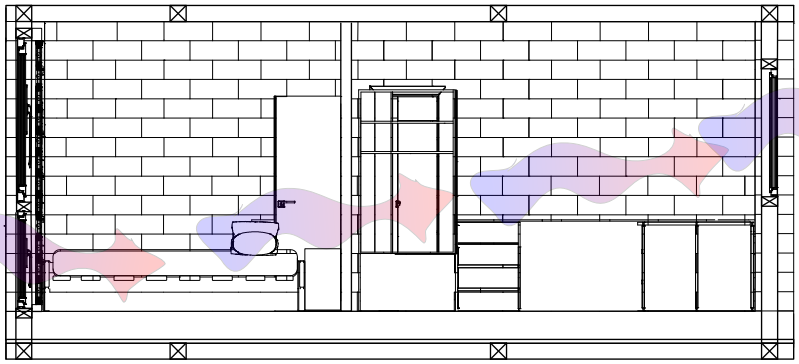


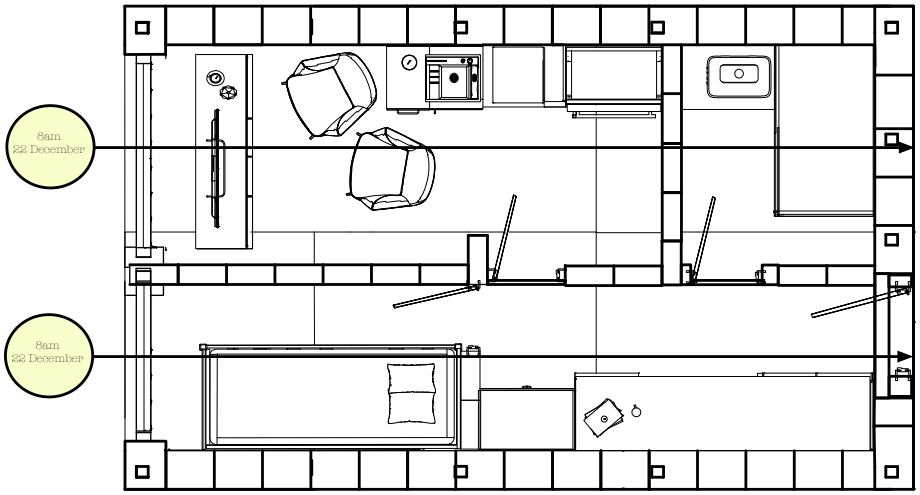


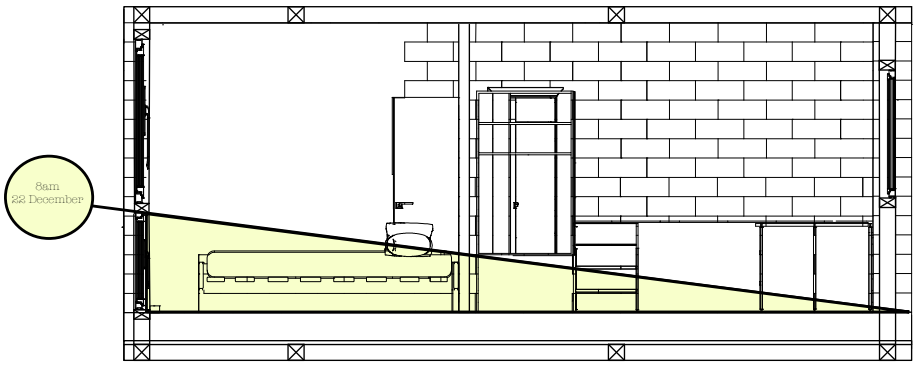


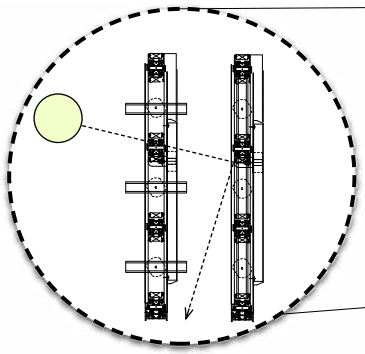
Two opening for cross ventilation lets air pass through the space freely. The closed space lets air travel above the inhabitants and has less effect on those using the space airing the unit and leaving with no issues from wind currents.

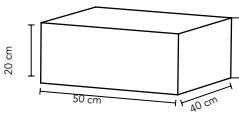





8am
22 December
8am
22 December


8am
22 December

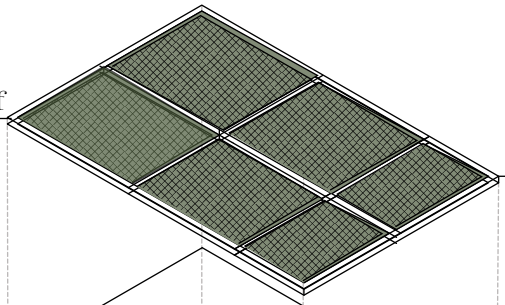




Thickness: 40cm	
Density (kg m^3)	330
Thermal conductivity (W/(mK)	0.07
Thermal transmittance U-vlaue (W/(m2K)	0.17
Resistance to water vapour (m)	4.50
Heat capacity (K/kgK)	1870
Sound absorption	0.8
Sound reduction	42.96

Green roof

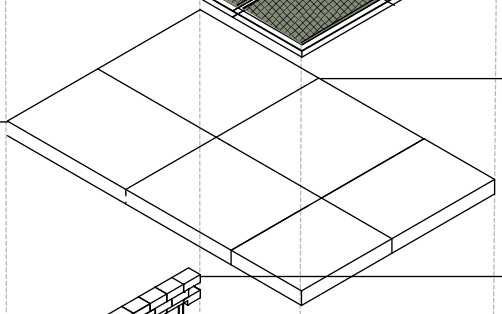
The roof of the top building has a green roof to catch water and ensure the hempcrete is not damaged, it can be moved according to the building project.



Green roof

Roof / Floor slabs

The floor slabs are made into panels and are lifted onto the floors placed on the structure, acting as roof and floor slabs. 12 slabs in total.



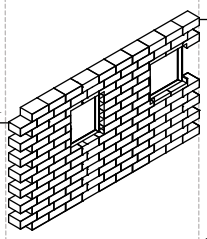
Floor slabs

1 .190cm x250cm
2 .298 x 250cm
3 .298 x 250cm

Wall Three

Back wall

The back of the house takes 140 bricks to construct- minus the bricks where the windows are placed. Both back windows are 1m by 1m.

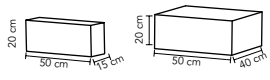


Internal Wall

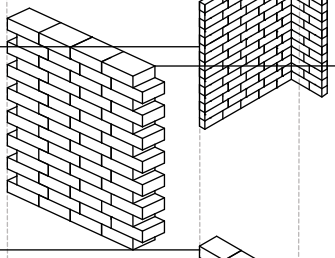
280 bricks

Internal Walls

The internal walls need no support and bricks can be layered and cut where doors are needed. 280 bricks are needed.



Internal / External

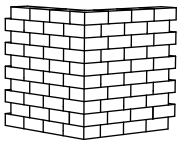


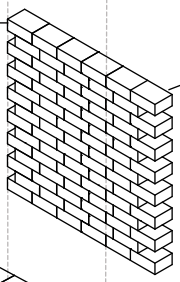
Wall Three

56 bricks

Walls

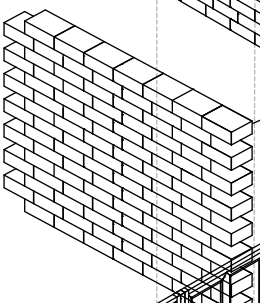
The wall is made from bricks using a stretcher bond there is no material waste and the only additions to the bricks would be to cut them to fit around the support pillars. The sides are the same so in total there are three walls but doubled up to make six in total for both sides. Taking 476 bricks to make both sides.





Wall Two

84 bricks



Wall One

98 bricks

Window

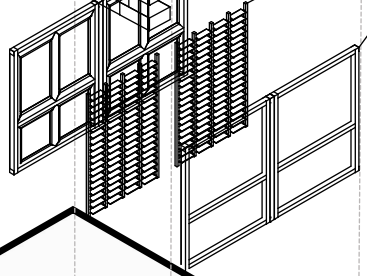
Louvers

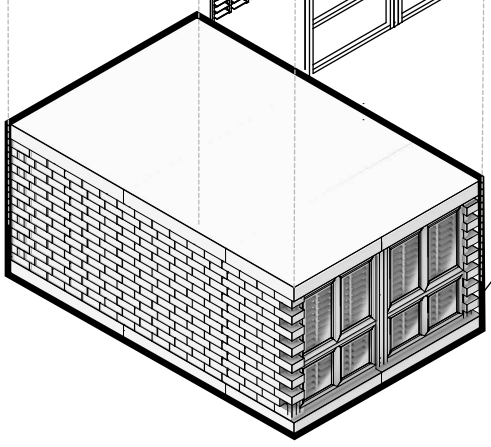
Frame

Lintel
Pillar

Window

The front part consists of: frame sealed by hempcrete, windows, and louvers to block out the sun annually. The frame uses a lintel to support the window which is attached to the support pillars. Together it forms the front part of the home.

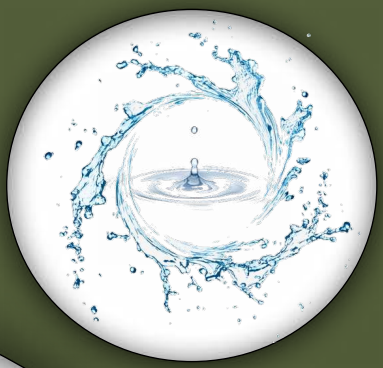




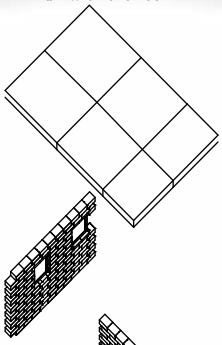
Housing

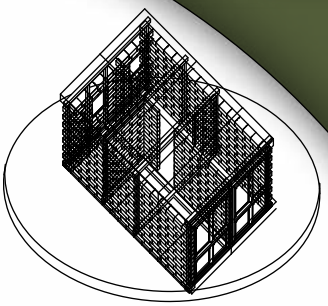


1. Mix binder with water

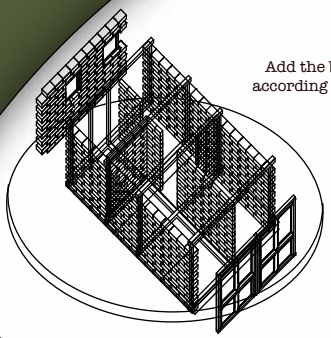


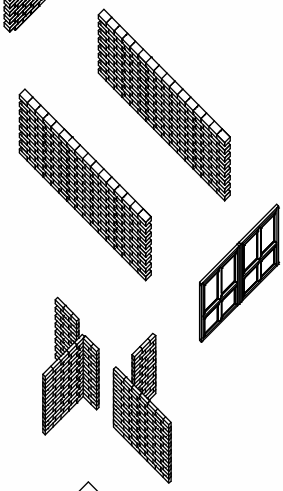
Finish with the roof

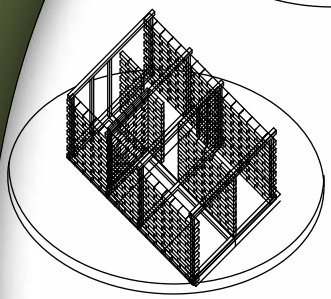




Add the back and front according to the catalogue

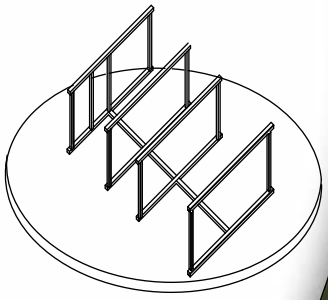




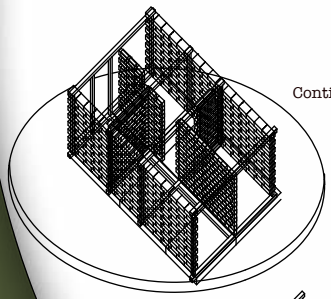


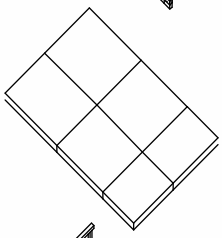
Finish and add lime render

Build strucrual frame

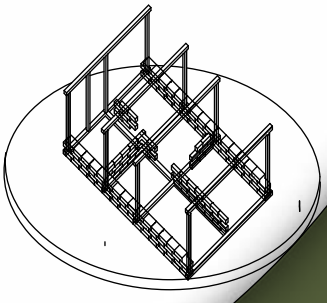


Contine to stack accroding to brick formation

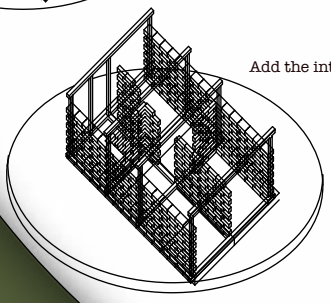


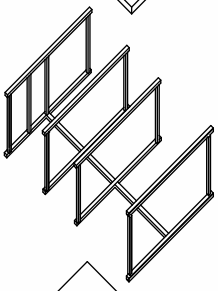


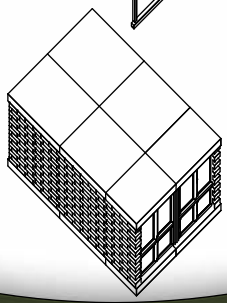
Add bricks between the fame



Add the internal bricks

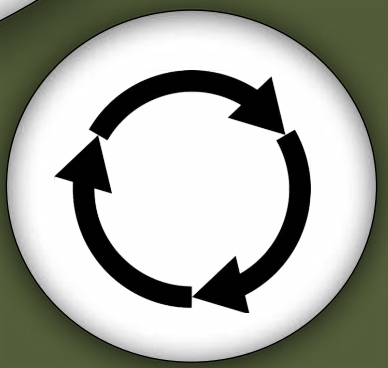








2. Continue to mix while adding the hemp



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2019 – 2020

This document summarises a modular hempcrete construction system developed through material testing, form finding and a full assembly catalogue. The bonded-brick wall, floor and roof system demonstrates a replicable, low-carbon approach to construction, and can serve as a reference for future natural-material building projects.



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